
CONFORMAL PARETO-GUIDED 3D MOLECULAR GENERATION: SHIFT-AWARE, CALIBRATED MULTI-CONSTRAINT SATISFACTION IN LOW-DATA

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ABSTRACT

De novo molecular generation for multi-parameter optimization often fails in the regime that matters most in early discovery: when only a few dozen to a few hundred labeled molecules are available per property, generators optimized against point-estimate predictors can exploit miscalibration and produce candidates that do not satisfy constraints under stronger, more structure-aware evaluation. This proposal targets reliability rather than predicted means by asking how to train a conditional generator to maximize the empirical rate of satisfying multiple hard property thresholds simultaneously, while preserving novelty and scaffold diversity, even as generation induces distribution shift away from the calibration data. We propose Conformal Pareto-Guided 3D Molecular Generation, which replaces point-reward optimization with calibrated feasibility margins derived from per-property conformal uncertainty objects, uses a shift monitor to trigger conservative recalibration when generated candidates drift out of distribution, and maintains a Pareto archive to avoid scalarization collapse across competing constraints. The planned experiments follow an iterative generate–score–archive–fine-tune loop with a preference-conditioned SMILES transformer generator, deep-ensemble property predictors, and split conformal calibration at a fixed nominal coverage, evaluated under a stronger conformer-ensemble interface built from RDKit ETKDG conformer generation and aggregation across multiple conformers per molecule. We will benchmark across multiple multi-property constraint bundles, three low-data regimes, and ten shared random splits, comparing against point-estimate optimization, uncertainty-penalized scalarization, conformal filtering without margin optimization, Pareto optimization without shift-aware recalibration, and 2D-only evaluation, with ablations isolating recalibration, Pareto archiving, diversity controls, and conformalization. Primary evaluation will report top-100 all-constraints satisfaction under the 3D conformer-ensemble evaluator, alongside conformal coverage on held-out test sets, the surrogate-to-evaluator gap, novelty and diversity metrics, and efficiency, providing a reproducible assessment of whether optimizing calibrated, shift-aware feasibility can materially improve multi-constraint success in low-data molecular design.

1 Introduction

Modern *de novo* molecular generation increasingly targets multi-parameter optimization, where candidates must satisfy several hard constraints (e.g., potency, selectivity, and safety) simultaneously rather than merely improve a single score. Yet a persistent failure mode remains: molecules optimized against learned property predictors often look excellent to the surrogate but fail when assessed by stronger, more structure-aware evaluators, especially when only a few hundred labeled molecules are available per property. This gap is particularly salient as the field moves toward 3D-aware generation and evaluation, where geometric validity and downstream structure-based assessments expose weaknesses of proxy objectives [1]. The core problem we address is therefore reliability: can a generator be trained to propose molecules that *actually* satisfy multiple constraints under low-data uncertainty, rather than exploiting point-estimate predictors?

This problem is important because practical discovery pipelines require a high hit rate of candidates that meet all criteria, not just improved predicted means. Uncertainty quantification is a natural lever: conformal prediction offers user-specified coverage guarantees and compound-specific uncertainty estimates that are widely used in QSAR decision-making and applicability-domain reasoning [2, 3]. If generation were driven by calibrated feasibility rather than raw

predictions, it could better align optimization with the operational question chemists care about: the probability that a molecule clears every threshold.

However, naive uncertainty-aware optimization is hard. Classical conformal guarantees rely on exchangeability assumptions that are stressed by non-i.i.d. conditions such as dataset shift [2]. Generative optimization intentionally induces such shift by pushing candidates away from the calibration distribution, precisely where predictors are most brittle. In addition, multi-constraint objectives can interact in ways that cause mode collapse, yielding narrow chemotypes that satisfy a scalarized reward but undermine novelty and downstream optimization flexibility.

These challenges have not been fully resolved by prior approaches. While diffusion-based 3D generators have advanced the ability to model molecular geometry, they also highlight practical issues such as the mismatch between proxy objectives and structure-based evaluation [1]. Recent target-aware, guided equivariant diffusion approaches further demonstrate how 3D generation can be steered toward binding-relevant objectives, increasing the need for reliable, calibrated constraint satisfaction when guidance is applied under limited labels [4]. Meanwhile, conformal prediction in cheminformatics is often used as a post hoc reporting or filtering tool rather than as the central optimization target [2]. Recent shift-aware conformal methods demonstrate that adaptive, individualized intervals can improve coverage-width trade-offs under OOD conditions [5], but they are typically studied in fixed prediction settings rather than inside a generation-time optimization loop.

We propose *Conformal Pareto-Guided 3D Molecular Generation*, a framework that makes calibrated multi-constraint satisfaction the objective. Our approach (i) optimizes conformalized feasibility margins using lower/upper confidence bounds relative to thresholds, (ii) incorporates shift-aware recalibration to remain conservative when generation departs from the calibration distribution, and (iii) evaluates candidates under a stronger 3D conformer-ensemble setting to directly measure the surrogate-to-evaluator gap emphasized in 3D generative modeling [1]. By integrating conformal uncertainty principles [2, 3] with adaptive shift-aware calibration ideas [5], we aim to improve the empirical rate of all-constraints satisfaction in the low-data regime without sacrificing novelty and diversity.

2 Related Work

3D molecular generative modeling with diffusion Diffusion models have become a dominant paradigm for *de novo* molecular generation, particularly for generating and reasoning over 3D molecular structures rather than only 2D graphs or strings. The recent review by Alakhdar *et al.* synthesizes how diffusion-based generators represent atoms/bonds, parameterize SE(3)-aware denoising networks, and condition generation on structural or property requirements, while also emphasizing evaluation practices and practical challenges specific to stable 3D generation (e.g., geometric validity and the gap between proxy objectives and downstream structure-based assessments) [1]. Relative to this line of work, our focus is not on proposing a new diffusion architecture, but on addressing a complementary failure mode that persists across generator families: when optimization is driven by point-estimate surrogates, generated molecules can exploit model weaknesses and fail under stronger (often more 3D-sensitive) evaluators. Thus, we build on the community’s move toward 3D-aware generation and evaluation highlighted in [1], but shift the optimization target from predicted means to calibrated, constraint-satisfaction margins intended to be robust under low-data uncertainty.

Conformal prediction for uncertainty quantification in cheminformatics Conformal prediction (CP) has been widely adopted in cheminformatics to provide user-specified coverage guarantees and compound-specific uncertainty estimates, with particular relevance to QSAR modeling, applicability-domain reasoning, and decision-making under limited labels [2]. However, as emphasized in this review, CP’s classical guarantees rely on exchangeability and can be stressed by non-exchangeable settings that are common in practice (e.g., dataset shift, deep learning pipelines, and other departures from i.i.d. assumptions) [2]. This limitation is central in generative design, where the model intentionally shifts the candidate distribution away from the calibration set. Our work is aligned with CP’s goal of calibrated, per-molecule uncertainty, but differs from much of the existing cheminformatics CP usage summarized in [2] in that we treat CP not merely as a post hoc reporting tool or screening filter; instead, we make conformalized feasibility margins the *optimization objective* for multi-constraint generation, and explicitly design mechanisms to remain conservative when generation induces distribution shift.

Shift-aware and adaptive conformal methods under OOD Recent work has begun to directly tackle the brittleness of uncertainty quantification under distribution shift by combining conformal calibration with mechanisms that adapt interval widths on a per-sample basis. Badkul and Xie propose an expert-routed conformal framework (TESSERA) that leverages mixture-of-experts diversity to produce *individualized* prediction intervals that maintain near-nominal coverage while improving the coverage-width trade-off under both i.i.d. and scaffold-based OOD splits in drug-discovery-relevant prediction tasks [5]. This is closely related in spirit to our shift-aware recalibration component: both approaches acknowledge that naive uncertainty estimates can become either invalid (under-cover) or uninformative

(overly wide) under shift, and aim to “right-size” uncertainty to track error under OOD conditions [5]. Our approach differs in problem setting and downstream use: rather than focusing solely on predictive UQ quality for a fixed supervised task, we integrate shift-aware conformalization into a *generation-time* loop and optimize multi-constraint satisfaction via conformal LCB/UCB margins. In doing so, we aim to translate adaptive, shift-robust uncertainty ideas like those in [5] into a practical mechanism that reduces reward hacking and improves empirical all-constraints satisfaction when candidates are re-evaluated by stronger (including 3D-sensitive) evaluators.

3 Method

Problem definition (multi-constraint, low-data, shift-prone generation). Let $\mathcal{X}$ denote the space of valid molecules (represented as SMILES/SELFIES strings or molecular graphs) and let K denote the number of properties. For each property $k \in \{1, \dots, K\}$, we assume an unknown target function $f_k : \mathcal{X} \rightarrow \mathbb{R}$ (regression) or $f_k : \mathcal{X} \rightarrow [0, 1]$ (probability of a positive class). Each property is associated with a constraint type and threshold(s) τ_k : (i) *must-be-high* constraints require $f_k(x) \geq \tau_k$, (ii) *must-be-low* constraints require $f_k(x) \leq \tau_k$, and (iii) *range* constraints require $f_k(x) \in [\tau_k^\ell, \tau_k^u]$. The goal is to learn a conditional generator G_θ that induces a distribution $q_\theta(x | w)$ over molecules given a preference vector $w \in \Delta^{K-1}$ (the probability simplex), such that samples from $q_\theta(\cdot | w)$ have high probability of satisfying *all* constraints simultaneously under a stronger evaluator, while remaining novel and diverse. A central difficulty is that the generator induces distribution shift relative to the labeled data used to fit predictors and calibrate uncertainty, which can invalidate naive uncertainty estimates under non-exchangeability [2].

Predictive models and conformalized uncertainty (why: calibrated feasibility rather than point estimates).

For each property k , we train a predictor $\hat{f}_k(x)$ and construct a conformal prediction object that returns a calibrated uncertainty region at nominal miscoverage level $\alpha \in (0, 1)$. Conformal prediction is used because it provides a principled, user-chosen coverage target that is widely used for compound-specific uncertainty in QSAR-style decision making [2, 3]. For regression properties, we use split conformal intervals of the form

$$\hat{C}_k(x) = [L_k(x), U_k(x)] \quad (1)$$

$$L_k(x) = \hat{f}_k(x) - \hat{q}_k, \quad U_k(x) = \hat{f}_k(x) + \hat{q}_k, \quad (2)$$

where $\hat{q}_k$ is a quantile of nonconformity scores computed on a held-out calibration set (e.g., absolute residuals). For classification-style properties (e.g., toxicity probability), we construct a conformal prediction set $\hat{S}_k(x) \subseteq \{0, 1\}$ using a suitable nonconformity score (e.g., based on predicted class probabilities), which yields a calibrated decision region rather than a single probability [2]. In both cases, the conformal object is treated as the interface that maps a molecule x to a conservative uncertainty summary used downstream for optimization.

Conformal feasibility margins (what: robust constraints; why: reduce reward hacking under low-data). We convert each constraint into a *robust feasibility margin* computed from conformal bounds, so that a molecule is attractive only if it clears thresholds with calibrated uncertainty rather than by exploiting overconfident point predictions. For each property k , define a scalar margin $m_k(x)$ as

$$m_k(x) = \begin{cases} L_k(x) - \tau_k, & \text{must-be-high} \\ \tau_k - U_k(x), & \text{must-be-low} \\ \min\{L_k(x) - \tau_k^\ell, \tau_k^u - U_k(x)\}, & \text{range (regression)} \end{cases} \quad (3)$$

and define the overall robust feasibility score as

$$m_{\min}(x) = \min_{k \in \{1, \dots, K\}} m_k(x). \quad (4)$$

A molecule is *robust-feasible* if $m_{\min}(x) \geq 0$. This construction directly operationalizes the decision-making perspective emphasized in conformal prediction for cheminformatics: rather than optimizing predicted means, we optimize a conservative proxy for the event that all constraints hold [2, 3]. The key design choice is to make the optimization target depend on (L_k, U_k) , because under low-data uncertainty, point-estimate optimization can select molecules in regions where predictors are brittle and miscalibrated, whereas conformalized margins explicitly penalize uncertainty by shrinking feasible regions when errors are likely [2].

Shift-aware recalibration during generation (what: detect shift and widen intervals; why: conformal validity can degrade under non-exchangeability). Classical conformal guarantees rely on exchangeability, which can be violated when the generator pushes candidates away from the calibration distribution [2]. To remain conservative

under generation-induced shift, we introduce a shift-aware recalibration layer that adapts interval widths (or switches to group-conditional calibration) when generated molecules appear out-of-distribution relative to the calibration set. Let $\phi(x) \in \mathbb{R}^d$ be an embedding of x (e.g., from a predictor’s penultimate layer). At generation time, we compare embeddings from a reference set $\mathcal{D}_{\text{cal}}$ to embeddings from a recent batch of generated candidates $\mathcal{B}_t$ using a two-sample discrepancy statistic $D(\phi(\mathcal{D}_{\text{cal}}), \phi(\mathcal{B}_t))$. When D exceeds a threshold, we apply a conservative inflation factor $\gamma_t \geq 1$ to widen regression intervals:

$$L_k^{(t)}(x) = \hat{f}_k(x) - \gamma_t \hat{q}_k, \quad (5)$$

$$U_k^{(t)}(x) = \hat{f}_k(x) + \gamma_t \hat{q}_k, \quad (6)$$

and recompute margins $m_k^{(t)}(x)$ accordingly. This mechanism is motivated by the observation that adaptive conformal methods can improve robustness of uncertainty under OOD conditions by tailoring interval widths to error patterns [5]. While our setting differs (generation-time optimization rather than fixed prediction), the design rationale is the same: when shift increases, uncertainty should become more conservative to avoid under-coverage and overconfident feasibility claims [5, 2].

Pareto-guided multi-constraint optimization (what: maintain a feasible frontier; why: avoid scalarization collapse). Optimizing $m_{\min}(x)$ alone can be overly conservative and can collapse to chemotypes that improve the single worst constraint at the expense of others. Instead, we treat the margin vector

$$\mathbf{m}(x) = (m_1(x), \dots, m_K(x)) \in \mathbb{R}^K \quad (7)$$

as a multi-objective signal and maintain an archive $\mathcal{A}$ of non-dominated candidates under Pareto dominance. We then train the generator to place probability mass on different regions of the Pareto frontier using preference conditioning. Given a sampled preference vector $w \in \Delta^{K-1}$, we define a linear utility

$$u_w(x) = w^\top \mathbf{m}(x), \quad (8)$$

and select target molecules from $\mathcal{A}$ that maximize $u_w(x)$ subject to diversity constraints (described below). The generator is updated to increase likelihood of these selected molecules under $q_\theta(\cdot \mid w)$, yielding a preference-conditioned generator that can trade off constraints without collapsing to a single scalarization. This design is motivated by the multi-parameter optimization setting emphasized in the introduction: satisfying multiple hard constraints is not equivalent to maximizing a single score, and Pareto-guided learning provides a structured way to preserve multiple feasible trade-offs rather than a single mode.

Diversity and novelty constraints (what: explicit constraints; why: preserve exploration and reduce degenerate solutions). To prevent mode collapse and trivial duplication, we incorporate explicit novelty and diversity constraints into archive maintenance and preference-conditioned selection. Let $s(x, x') \in [0, 1]$ denote a similarity measure (e.g., Tanimoto similarity on fingerprints). We enforce (i) a novelty constraint that rejects candidates with $\max_{x' \in \mathcal{D}_{\text{train}}} s(x, x')$ above a threshold, and (ii) a diversity constraint that limits the number of selected molecules per scaffold group. These constraints are applied as hard filters or as penalties in the selection objective:

$$\tilde{u}_w(x) = u_w(x) - \lambda \cdot \text{sim_penalty}(x; \mathcal{S}), \quad (9)$$

where $\mathcal{S}$ is the current selected set and sim_penalty increases with similarity to already-selected molecules. The rationale is that multi-objective optimization can otherwise concentrate on narrow chemotypes that satisfy the surrogate constraints but provide limited value for downstream discovery; explicit diversity constraints preserve a broader set of feasible hypotheses for subsequent refinement.

3D conformer-ensemble evaluation interface (what: evaluate with 3D sensitivity; why: measure and reduce the surrogate-to-evaluator gap). Many downstream criteria are sensitive to 3D geometry and conformational variability, and 3D generative modeling work highlights the importance of evaluation beyond simple 2D proxy objectives [1]. We therefore define a 3D conformer-ensemble evaluation interface $E(x)$ that maps a molecule to an evaluator score (or constraint outcomes) by aggregating over a set of conformers $\{c_j(x)\}_{j=1}^J$:

$$E(x) = \text{Agg}(e(x, c_1(x)), \dots, e(x, c_J(x))), \quad (10)$$

where $e(\cdot)$ is a conformer-level scoring function and $\text{Agg}(\cdot)$ is an aggregation operator (e.g., min/mean/max). This evaluator is used to assess whether robust-feasible molecules under conformal margins remain feasible under a stronger, more geometry-aware check, directly targeting the proxy-to-evaluator mismatch emphasized in 3D generation and evaluation [1]. Importantly, the evaluator is treated as an external assessment module: the core optimization signal remains the conformal margin vector $\mathbf{m}(x)$, while $E(x)$ is used to quantify reliability of generated candidates under a stricter standard.

Overall algorithm (summary). Conformal Pareto-Guided 3D Molecular Generation combines (i) per-property conformal uncertainty objects to compute calibrated bounds [2, 3], (ii) robust feasibility margins that transform bounds into constraint-satisfaction objectives, (iii) shift-aware recalibration that becomes more conservative under generation-induced OOD behavior [5, 2], (iv) Pareto-archive preference conditioning to avoid scalarization collapse, and (v) a 3D conformer-ensemble evaluator to measure the surrogate-to-evaluator gap highlighted in 3D generative modeling [1]. Together, these components define a generation-time loop that optimizes for calibrated multi-constraint satisfaction rather than point-estimate rewards, with explicit mechanisms to remain conservative under shift and to preserve diverse feasible trade-offs.

4 Experimental Setup

4.1 Overview and experimental factors

All experiments follow the same generate–score–archive–fine-tune loop described in Section 3, with the optimization signal given by conformal feasibility margins and the evaluation performed under a stronger conformer-ensemble interface to quantify the surrogate-to-evaluator gap emphasized in 3D molecular generation and evaluation [1]. We vary three primary experimental factors: (i) low-data regime ($n \in \{50, 100, 500\}$ labeled molecules per property), (ii) multi-property setting (at least three distinct constraint bundles), and (iii) random split seed (10 seeds shared across all methods). For each configuration, we train predictors, calibrate conformal uncertainty, run the generation loop for a fixed number of rounds, and evaluate the top-ranked molecules under the 3D conformer-ensemble evaluator.

4.2 Datasets, task construction, and preprocessing

Property datasets and rationale. We construct multi-constraint design settings by combining commonly used molecular property prediction datasets spanning activity-like regression tasks and ADMET-like proxies, because multi-parameter optimization is the practical regime where point-estimate surrogate optimization is most vulnerable to miscalibration and reward hacking under limited labels [2]. Each setting includes $K \geq 3$ properties with explicit constraint types (must-be-high, must-be-low, or range) and fixed thresholds $\{\tau_k\}_{k=1}^K$ chosen *a priori* per setting.

Molecule standardization. All molecules are standardized prior to splitting: RDKit sanitization, salt/fragment removal, canonicalization to a unique SMILES string, and deduplication across the full labeled pool to prevent leakage between train/calibration/test. We compute Bemis–Murcko scaffolds for each molecule to support scaffold-based grouping for diversity metrics and (when enabled) group-conditional conformal calibration.

Low-data splits. For each property dataset, we create 10 random splits with a fixed seed list shared across methods. Within each split and each low-data regime n , we subsample n labeled molecules for supervised training (stratified for classification properties), allocate a calibration set from the remaining labeled molecules (targeting 20% of the remaining pool, with a minimum size when available), and reserve the rest as a held-out test set used only for predictor evaluation and conformal coverage reporting. This split-conformal protocol is used because conformal prediction in cheminformatics is commonly operationalized via a held-out calibration set to obtain user-specified marginal coverage targets [2, 3].

4.3 Models and implementation details

Generator. We use a SMILES-based autoregressive transformer decoder as the generator G_θ , optionally with SELFIES tokenization to improve syntactic validity. The generator is preference-conditioned by a weight vector $w \in \Delta^{K-1}$, implemented either by (i) concatenating a learned embedding of w to token embeddings at each position or (ii) prepending a short sequence of learned “preference tokens” derived from w . The generator is initialized from a generic molecular language model pretrained on a large unlabeled corpus (e.g., ZINC/ChEMBL-like) using next-token prediction, then fine-tuned on the task-specific training molecules before entering the iterative optimization loop.

Property predictors and uncertainty. For each property k , we train an independent predictor $\hat{f}_k$ using either a graph neural network (e.g., MPNN/GIN) or a molecular transformer encoder. To stabilize uncertainty estimates in low-data, we use a deep ensemble of size $M = 5$ with different random initializations and bootstrap resampling of the training set. Regression predictors are trained with mean squared error; classification predictors are trained with binary cross-entropy. Early stopping is performed using a validation subset carved from the training set (distinct from the conformal calibration set) to avoid tuning on calibration labels.

Conformal calibration. We use split conformal prediction at nominal coverage $1 - \alpha = 0.9$ to construct per-property uncertainty objects, consistent with conformal prediction practice in cheminformatics for producing compound-specific uncertainty with user-chosen coverage [2, 3]. For regression, the nonconformity score is the absolute residual on the calibration set; the conformal radius $\hat{q}_k$ is the $(1 - \alpha)$ quantile of calibration scores, yielding intervals $[L_k(x), U_k(x)]$. For classification-style constraints, we construct conformal prediction sets using a probability-based nonconformity score (e.g., $1 - \hat{p}(y | x)$) so that the downstream constraint can be expressed in terms of conservative set membership rather than a single probability estimate [2].

Shift-aware recalibration. Because conformal validity can degrade under non-exchangeability and dataset shift [2], we implement a shift monitor and conservative recalibration mechanism inspired by adaptive conformal ideas that tailor uncertainty under OOD conditions [5]. Concretely, at each generation round t , we compute embeddings $\phi(x)$ for calibration molecules and for a recent batch of generated candidates, then compute a two-sample discrepancy statistic using maximum mean discrepancy (MMD) with an RBF kernel. If the shift statistic exceeds a preset threshold or yields a p-value below 0.01, we inflate regression intervals by a factor $\gamma_t \in [1, 2]$ and recompute feasibility margins. In addition, we optionally switch to scaffold-binned (group-conditional) conformal calibration when scaffold bins have sufficient calibration support, falling back to global conformal otherwise.

4.4 Generation loop, archive maintenance, and fine-tuning protocol

Per-round candidate generation. For each split and low-data regime, we run $T = 10$ optimization rounds. In each round, we sample $N = 20,000$ molecules from the current generator using nucleus sampling ($\text{top-}p \in \{0.9, 0.95\}$) and temperature ($\in \{0.8, 1.0\}$), with a maximum sequence length of 128 tokens. We discard invalid molecules using RDKit parsing and sanitization, and we remove duplicates within the round and against the existing archive.

Scoring with conformal feasibility margins. For each valid molecule x , we compute per-property conformal bounds (after any shift-aware recalibration) and then compute the margin vector $\mathbf{m}(x)$ and the scalar worst-constraint margin $m_{\min}(x)$ as defined in Section 3. For range constraints, we compute the margin as the minimum of the lower-bound and upper-bound slack relative to the allowed interval. These margins are the only signals used for selecting molecules to add to the Pareto archive and for constructing preference-conditioned training pairs.

Pareto archive update. We maintain an archive $\mathcal{A}$ of up to 50,000 molecules, storing for each entry: SMILES, margin vector $\mathbf{m}(x)$, novelty/diversity metadata, and (when computed) 3D evaluator outcomes. At each round, we add up to 2,000 new molecules selected by Pareto dominance on $\mathbf{m}(x)$ with an optional ϵ -dominance tolerance, then prune the archive to enforce the maximum size while preserving frontier diversity (e.g., by keeping a spread across preference directions).

Diversity and novelty controls. To preserve exploration and avoid degenerate duplication, we enforce three controls during archive insertion and top- N selection: (i) novelty filter rejecting molecules whose maximum Tanimoto similarity to any training molecule exceeds 0.6 (computed on Morgan fingerprints), (ii) scaffold quota limiting to at most 5 molecules per Bemis–Murcko scaffold in the top-100 set used for reporting, and (iii) an intra-batch similarity penalty during selection for fine-tuning, proportional to the mean pairwise similarity among selected molecules.

Preference-conditioned fine-tuning. After updating $\mathcal{A}$, we fine-tune the generator for one epoch using either maximum likelihood on selected exemplars or a preference-optimization objective. For each minibatch, we sample a preference vector $w \sim \text{Dirichlet}(\mathbf{1})$, select a positive set of molecules maximizing $u_w(x) = w^\top \mathbf{m}(x)$ subject to diversity constraints, and select negatives either uniformly from the archive or from near-threshold violators. We optimize with AdamW using learning rate 10^{-4} and batch size 64.

4.5 3D conformer-ensemble evaluator

Conformer generation. To evaluate candidates under a stronger, geometry-aware interface, we generate a conformer ensemble for each molecule using RDKit ETKDG (v3) with $J \in \{10, 20\}$ conformers per molecule, followed by force-field minimization (UFF or MMFF94). This design follows the principle that 3D generative modeling should be assessed with geometry-sensitive checks rather than only 2D proxy scores [1].

Ensemble aggregation and evaluator outputs. For each conformer $c_j(x)$, we compute conformer-level scores $e(x, c_j(x))$ for each constraint in the evaluator, then aggregate across conformers using $\text{Agg} \in \{\min, \text{mean}, \max\}$ depending on the constraint semantics (e.g., worst-case aggregation for “must-be-low” risk proxies). The evaluator

returns (i) per-constraint pass/fail outcomes and (ii) an overall all-constraints pass/fail indicator used for the primary endpoint. The evaluator is used for reporting and periodic auditing of the archive, not as a direct training reward, to keep the optimization target aligned with calibrated feasibility margins while still measuring the surrogate-to-evaluator gap highlighted in 3D evaluation discussions [1].

4.6 Baselines and ablations

Baselines. We compare against the following baselines, each run with the same data splits, generator backbone, sampling budget, and number of rounds:

1. **Point-estimate optimization:** optimize a scalar reward built from predicted means relative to thresholds (no conformalization).
2. **Scalarized multi-objective without conformal:** optimize $w^\top(\hat{\mu}(x) - \tau)$ with an uncertainty heuristic penalty (e.g., mean minus β times ensemble standard deviation).
3. **Conformal filter-only:** generate using point-estimate optimization, then filter candidates post hoc by robust-feasible status $m_{\min}(x) \geq 0$.
4. **Pareto without shift-aware recalibration:** optimize conformal margins and maintain a Pareto archive, but fix $\gamma_t = 1$ and disable group-conditional calibration.
5. **2D-only evaluation:** report success using only 2D predictor-based feasibility rather than the 3D conformer-ensemble evaluator to quantify evaluator mismatch [1].

Ablation studies. To isolate contributions of each component, we run ablations that remove: (i) shift-aware recalibration, (ii) Pareto archive (replace with scalar $m_{\min}(x)$ optimization), (iii) diversity controls, and (iv) conformalization (means only). These ablations directly test the method’s design motivations: conformalization for calibrated feasibility [2, 3], shift-aware conservatism under OOD [5, 2], and 3D-sensitive evaluation [1].

4.7 Evaluation metrics and statistical protocol

Primary endpoint. The primary endpoint is the **top-100 all-constraints satisfaction rate under the 3D conformer-ensemble evaluator**, defined as the fraction of the 100 highest-ranked generated molecules (ranked by $m_{\min}(x)$, with ties broken by $w^\top \mathbf{m}(x)$ under a fixed reporting preference) that satisfy all evaluator constraints.

Secondary metrics. We report:

1. **Conformal coverage:** empirical marginal coverage on the held-out test set for each property at nominal 90% coverage, reported per property and averaged, reflecting conformal prediction’s user-specified coverage objective [2, 3].
2. **Surrogate-to-evaluator gap:** difference between the predicted robust-feasible rate (by conformal margins) and the evaluator-feasible rate under the 3D conformer ensemble, motivated by the proxy-to-evaluator mismatch emphasized in 3D generation evaluation [1].
3. **Novelty:** mean Tanimoto distance to the training set and the fraction of molecules passing the novelty filter.
4. **Diversity:** number of unique Bemis–Murcko scaffolds in the top-100 and internal diversity computed as $1 - \text{mean pairwise similarity}$.
5. **Pareto quality:** hypervolume and frontier size computed on the margin vectors $\mathbf{m}(x)$.
6. **Efficiency:** wall-clock time per round and conformer-generation time per molecule (reported descriptively to support reproducibility).

Statistical testing across splits. All metrics are aggregated across the 10 random splits. For the primary endpoint, we use paired bootstrap resampling over splits to compute 95% confidence intervals for differences between our method and each baseline, and we report two-sided p-values from the bootstrap distribution. We treat each split as the unit of replication to reflect variability due to low-data sampling and model initialization.

4.8 Reproducibility checklist

To enable replication, we will release: (i) the fixed list of split seeds and the exact train/calibration/test indices per split, (ii) model configurations and hyperparameters for generator, predictors, and conformal calibration, (iii) the full list of

generated SMILES per round per method, (iv) conformal calibration quantiles and shift statistics per round, and (v) scripts for computing all metrics including the conformer-ensemble evaluator interface. This emphasis on transparent uncertainty calibration and evaluation is aligned with the role of conformal prediction as a decision-support tool in cheminformatics [2, 3] and with the broader push toward rigorous 3D-aware evaluation in molecular generation [1].

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